DE–504 **11**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2014.

ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Prove: If G is a finite group and $a \in G$, then $= \circ(a) | \circ(G)$. (10)

   (b) Prove: If ϕ is a homomorphism of G into $\overline{G}$ with kernel K, then K is a normal subgroup of G. (10)

2. (a) Prove: Let G be a group and ϕ an automorphism of G. If a $\in G$ is of order $= o(a) > 0$ then $o(\phi(a)) = o(a)$. (10)

   (b) Prove : S_{p^k} has a p-Sylow subgroup. (10)

3. (a) Prove: Let R be a commutative ring with unit element whose only ideals are (0) and R itself. Then show that R is a field. (10)

   (b) (i) Prove: An Euclidean ring possesses a unit element. (5)

   (ii) Prove: Let R be an Euclidean ring. Then any two elements a and b in R have a greatest common divisor d. Moreover $d = \lambda a + \mu b$, for some $\lambda, \mu \in R$. (5)

4. (a) Prove: If p is a prime number of the form 4n + 1, then we can solve the congruence $x^2 \equiv -1 \bmod p$.(10)

(b) The ideal A = (a_o) is a maximal ideal of the Euclidean ring R, if and only if a_o is a prime element of R. (10)

5. (a) Prove : If $v_1, \cdots, v_n$ is a basis for V over F and $w_1 \cdots, w_m$ in V are linearly independent over F, then $m \le n$. (10)

(b) Prove: Let V be a finite -dimensional inner product space; then V has an orthonormal set as a basis. (10)

6. (a) Prove: There exists subspace W of V, invariant under T, such that if $V = V_1 \oplus W$. (10)

(b) Prove: $T \in A(V)$ is unitary if and only if $T\,T^* = 1$. (10)

7. (a) Prove : The polynomial $f(x) \in F[x]$ has multiple roots. If and only if $f(x)$ and $f'(x)$ have a nontrivial roots (that is, of positive degree) common factor. (10)

(b) Prove : Let K be a normal extension of F and let H be a subgroup of $G(K,F)$; Let $K_H = \{x \in K \mid \sigma(x) = \text{x}$ for all $\sigma \in H\}$ be the fixed field of H. Then

(i) $[K : K_H] = o(H)$.

(ii) $H = G(K, K_H)$. In particular, when $H = G(K,H)$, $[K : F] = o(G(K,F)))$. (10)

8. Prove : If K is a finite extension of F, then $G(K,F)$ is a finite group and its order, $\circ(G(K,F))$ satisfies the inequality $\circ(G(K,F)) \le [K : F]$. (20)

DE–505 **12**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2014.

REAL ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Prove every non-empty open set S in R is the union of a countable collection of disjoint component intervals of S. (10)

   (b) State and prove Heine-Borel covering theorem. (10)

2. Let $f : S \to T$ be a function from one metric space (S, d_s) to another (T, d_T). Let A be a compact subset of S and assume that f is continuous on A. Prove that f is uniformly continuous on A. (20)

3. (a) State and prove 'intermediate –value theorem' for derivatives. (10)

   (b) State and prove '*Taylor's formula*' with remainder. (10)

4. State and prove '*implicit function theorem*'. (20)

5. (a) State and prove 'Bounded Convergence Theorem'. (10)

(b) Let $\propto$ be of bounded variation on $[a,b]$. Assume that each term of the sequence $\{f_n\}$ is a real valued function such that $f_n \in R(\propto)$ on $[a,b]$ and define $g_n(x) = \int_a^x f_n(t)\, d\propto(t)$. Prove that

(i) $f \in R(\propto)$ on $[a,b]$

(ii) $g_n \to g$ uniform on $[a,b]$, where $g(x) = \int_a^x f(t)\, d\propto(t)$. (10)

6. (a) Prove : The sequence of functions $\{f_n\}$, defined on E, converges uniformly on E if and only if for every $\varepsilon > 0$ there exists an integer N such that $m \geq N, n \geq N, x \varepsilon E$ implies. $|f_n(x) - f_m(x)| \leq \varepsilon$. (10)

(b) Prove : Suppose $\{f_n\}$ is a sequence of functions, differentiable on $[a,b]$ and such that $\{f_n(x_0)\}$ converges for some point x_0 on $[a,b]$. Then $\{f_n\}$ converges uniformly on $[a,b]$, to a function f, and $f'(x) = \lim_{n\to\infty} \int_n^{'}(x), (a \leq x \leq b)$. (10)

7. (a) State and prove *'Lebesque dominated* convergence theorem*'*. (10)

(b) State and prove *'Egoroffs* theorem'. (10)

8. (a) Let $\langle f_n \rangle$ be a sequence of measureable functions that convergences in measure to f. Prove that there exists a subsequence $\langle f_{nk} \rangle$ that converges to f almost every-where. (10)

(b) State and prove 'Lusin's theorem'. (10)

DE–506 **13**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2014.

DIFFERENTIAL EQUATIONS AND NUMERICAL METHODS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) State and prove the Existence theorem.

   (b) Find two linearly independent solutions of the equation $(3x-1)^2 y'' + (9x-3)y' - 9y = 0$.

2. Derive Legendre differential equation.

3. (a) Find the solution of the equation $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = x - y$.

   (b) Solve $p^2 x + q^2 y = z$.

4. (a) Find the steady state temperature at any point of a square plate whose two adjacent sides are kept at the constant temperature 100°C.

   (b) Reduce the equation $\frac{\partial^2 z}{\partial x^2} + x^2 \frac{\partial^2 z}{\partial y^2} = 0$ to canonical form.

5. (a) Using Newton's method, find a root of the equation $e^x = 4x$, which is approximately, correct to three places parts of decimal.

   (b) Using Chebyshev polynomials obtain the least squares approximation of second degree for $f(x) = x^4$ on [–1, 1].

6. Calculate the polynomial of degree ≤ 3 which minimizes $\int_{-1}^{1} [e^x - P(x)]^2 dx$ over all polynomials $P(x)$ of degree ≤ 3.

7. Let $P_n(x)$ be the polynomial of degree $\leq n$ which interpolates $f(x)$ at the points $x_0 < x_1 < x_2 ... < x_n$ in $[a, b]$. Prove that the Lagrange polynomial is given by $\prod_{j=0}^{n} \left(\frac{x - x_j}{x_i - x_j} \right)$ for $i \neq j$.

8. Find the solution of the differential equation $xy' = x - y$, $y(x) = 2$ at $x = 2.1$ correct to five decimal places, using

   (a) Taylor's method and

   (b) Euler's method.

DE–507 **14**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2014.

OPERATIONS RESEARCH

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Show by simplex method that the following LPP has infinite number of non-basic feasible optimal solutions. (10)

Maximize $z = 4x_1 + 10x_2$

Subject to :

$$\begin{aligned} 2x_1 + x_2 &\le 10 \\ 2x_1 + 5x_2 &\le 20 \\ 2x_1 + 3x_2 &\le 18 \\ x_1, x_2 &\ge 0. \end{aligned}$$

(b) Use dual simplex method to solve the LPP. (10)

Minimize $z = 3x_1 + 2x_2$

Subject to :

$$\begin{aligned} 3x_1 + x_2 &\ge 3 \\ 4x_1 + 3x_2 &\ge 6 \\ x_1, x_2 &\ge 3 \\ x_1, x_2 &\ge 0. \end{aligned}$$

2. (a) Explain Branch-and-Bound (B&B) Algorithm in details. (10)

(b) Consider the problem.

Maximize $z = 45x_1 + 100x_2 + 30x_3 + 50x_4$

Subject to :

$$7x_1 + 10x_2 + 4x_3 + 9x_4 \leq 1{,}200$$
$$3x_1 + 40x_2 + x_3 + x_4 \leq 1{,}800$$
$$x_1, x_2, x_3, x_4 \geq 0.$$

If a new variable, x_7 is added to this problem with a column $\begin{bmatrix} 10 \\ 10 \end{bmatrix}$ and $c_7 = 120$, find the change in the optimal solution. (10)

3. (a) A farmer wants to decide which of the three crops he should plant on his 100-acre farm. The profit from each is dependent on the rainfall during the growing season. The farmer has categorized the amount of rainfall as high, medium and low. His estimated profit for each is shown in the table below :

Rainfall	Estimated conditional profit (Rs.)		
	Crop A	Crop B	Crop C
High	8000	3500	5000
Medium	4500	4500	5000
Low	2000	5000	4000

If the farmer wishes to plant only one crop, decide which should be his best crop using,

(i) Maximax criterion

(ii) Maximin criterion

(iii) Hurwicz criterion (farmer's degree of optimism being 0.6)

(iv) Laplace criterion

(v) Minimax regret criterion. (10)

(b) Use dynamic programming to solve the LPP : (10)

Maximize $z = 3x_1 + 5x_2$

Subject to constraints :

$$x_1 \leq 4$$
$$x_2 \leq 6$$
$$3x_1 + 2x_2 \leq 18 \text{ and}$$
$$x_1, x_2 \geq 0.$$

4. (a) Using dominance property solve the following game : (10)

$$\begin{array}{cccc} \text{I} & \text{II} & \text{III} & \text{IV} \\ \end{array}$$
$$\begin{pmatrix} 3 & 2 & 4 & 0 \\ 3 & 4 & 2 & 4 \\ 4 & 2 & 4 & 0 \\ 0 & 4 & 0 & 8 \end{pmatrix}$$

(b) Solve the following game by linear programming.

	B_1	B_2	B_3	Row min
A_1	3	–1	–3	–3
A_2	–2	4	–1	–2
A_3	–5	–6	2	–6
Column max	3	4	2	

The value of the game v ,lies between –2 and 2. (10)

5. (a) Find the critical path and the expected time of the project from the following :

Activity	Predecessors	Estd. time (week)
A	–	2
B	A	3
C	A	4
D	B, C	6
E	–	2
F	E	8

Also calculate the slack for each activity. (10)

(b) Explain PERT calculation in details. (10)

6. (a) Explain the different types of inventory models. (10)

(b) Neon light of the U of A campus are replaced at the rate of 100 units per day. The physical plant orders the neon lights periodically. It costs Rs. 100 to initiate a purchase order. A neon light kept in storage is estimated to cost amount Rs. .02 per day. The lead time between placing and receiving an order is 12 days. Determine the optimal inventory policy for ordering the neon lights. (10)

7. (a) In a railway marshalling yard, goods, trains arrive at a rate of 30 trains per day. Assuming that the inter-arrival time follows an exponential distribution and the service time distribution is also exponential with an average 36 minutes. Calculate the following : (10)

(i) The mean queue size

(ii) The probability that the queue size exceeds 10. If the input of trains increase to an average 33 per day, what will be the change in (i) and (ii)?

(b) Explain the role of exponential distribution in queueing theory. (10)

8. (a) Explain the queueing model $(M/M/1):(GD/\infty/\infty)$. (10)

(b) Explain the queueing model $(M/M/\infty):(GD/\infty/\infty)$. (10)

DE–509 **21**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2014.

COMPLEX ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) State and prove the C-R equations for the functions z^2 and z^3. (10)

   (b) State and prove Abel's limit theorem. (10)

2. (a) Prove : If Q is a polynomial of degree n with distinct roots $\alpha_1 . \alpha_2 \ldots \alpha_n$ and if P is a polynomial of degree $< n$, show that $\frac{P(z)}{Q(z)} = \sum_{k=1}^{n} \frac{P(\alpha_k)}{Q'(\alpha_k)(z-\alpha_k)}$. (10)

   (b) Discuss the transformation $w = \sin z$. (10)

3. (a) State and prove Cauchy's theorem for a rectangle. (10)

   (b) Prove : Suppose that $f(z)$ is analytic in an open disk Δ, and let γ be a closed curve in Δ. For any point a not on γ, $n(\gamma, a) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)\, dz}{z-a}$, where $n(\gamma, a)$ is the index of with respect to γ. (10)

4. (a) Prove : If $f(z)$ is analytic and non-constant in a region Ω, then its absolute value $|f(z)|$ has no maximum in Ω. (The maximum principle). (10)

(b) State and prove Morera's theorem. (10)

5. (a) State and prove Cauchy's residue theorem. (10)

(b) Prove : Let $\gamma \sim 0$ in Ω and *f, g* be analytic in Ω and $n(\gamma, a) = 0$ or 1 for points $a \notin \gamma$. Then $|f(z) - g(z)| < |g(z)|$ on γ. (Note the strict inequality implies *f* and *g* have the same number of zeros enclosed by γ. (10)

6. (a) Prove that a function which is meromorphic in the whole plane is the quotient of two entire functions. (10)

(b) State and prove Jensen's formula. (10)

7. (a) Prove : A discrete module consists either of zero alone, of the integral multiples $n\omega$ of a single complex number $\omega \neq 0$, or of all linear combinations $n_1\omega_1 + n_2\omega_2$ with integral coefficients of two numbers ω_1, ω_2 with non-real ratio ω_1 / ω_2. (10)

(b) Prove : The sum of the residues of an elliptic function is zero. (10)

8. (a) Prove : An elliptic function without poles is a constant. (10)

(b) Explain the Weierstrass $\wp$-function. (10)

DE–510 **22**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2014.

TOPOLOGY AND FUNCTIONAL ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) State and prove Kuratowski closure axioms. (10)

   (b) Let X be a topological space. Then prove that :

   (i) any intersection of closed sets in X is closed; and

   (ii) any finite union of closed sets in X is closed. (10)

2. (a) State and prove Tychonoff's theorem. (10)

   (b) Prove that a metric space is sequentially compact $\Leftrightarrow$ it has the Bolzano-Weierstrass property. (10)

3. (a) Show that every compact Hausdorff's space is normal. (10)

   (b) In a Hausdorff space prove that, any point and disjoint compact subspace can be separated by open sets, in the sense that they have disjoint neighborhoods. (10)

4. Prove that locally compact Hausdorff spaces are compact. (20)

5. (a) (i) Show that every unit sphere is convex. (5)

(ii) Show that the parallelogram law is not true in l^n. (5)

(b) Prove : Let M be a linear subspace of a normal linear space N, and let f be a functional defined on M. If x_0 is a vector not in M, and if $M_0 = M + x_0$ is the linear subspace. Spanned by M and x_0, then f can be extended to a functional f_0 defined on M_0 such that $\|f_0\| = \|f\|$. (10)

6. (a) If T is a linear transformation from N into itself, then prove that $\|T^*\| = \|T\|$, where T^* is the conjugate of T. (10)

(b) State and prove the 'uniform Boundedness theorem'. (10)

7. (a) Prove : If M is a proper closed linear subspace of a Hilbert space H, then there exists a non-zero vector z_0 in H such that $z_0 \perp M$. (10)

(b) Discuss the Gram-Schmidt orthogonalization process. (10)

8. (a) If T be an operator on H such that $(Tx, x) = 0$ for all x, then prove that $T = 0$. (6)

(b) Prove that an operator T on H is unitary $\Leftrightarrow$ it is an isometric isomorphism of H onto itself. (7)

(c) Prove : If N is a normal operator on H, then $\|N^2\| = \|N\|^2$. (7)

DE–511 **23**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2014.

GRAPH THEORY

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Show that if there is a (u, v) – walk in G, then there is also $a(u, v)$ path in G. (10)

   (b) Prove that every circuit has an even number of edges in common with any cut-set. (10)

2. (a) Prove that the maximum vertex connectivity one can achieve with a graph G of n vertices and e edges $(e \geq n-1)$ is the integral part of the number $2e/n$. (10)

   (b) State and prove Sperner's lemma. (10)

3. (a) Define closure of the graph G. Also show that the closure of G is well defined. (10)

   (b) Prove : A weak digraph D is Eulerian iff every point of D has equal indegree and outdegree. (10)

4. (a) Prove the following statements are equivalent for a connected graph G.

   (i) G is Eulerian

   (ii) Every point of G has even degree

   (iii) The set of edges of G can be partitioned into cycles. (10)

   (b) If G is a simple graph with $\gamma \geq 3$ and $\delta \geq \gamma / 2$, prove that G is Hamiltonian. (10)

5. (a) Prove the following :

   (i) If G is $k-$critical, then $\delta \geq k-1$.

   (ii) In a critical path, no vertex cut is a clique. (10)

   (b) (i) If G is not connected, prove that $\overline{G}$ is connected.

   (ii) Show that a connected graph with n vertices and n-1 edges is a tree. (10)

6. (a) Define 'Ramsey numbers' and also state and prove Ramsey theorem. (10)

   (b) Prove :

   (i) $P_{k_n}(\lambda) = \lambda(\lambda-1)(\lambda-2).....(\lambda-n+1)$.

   (ii) Any tree T on n vertices has a chromatic polynomial $\lambda(\lambda-1)^{n-1}$. (10)

7. (a) Prove that δ = 0 for a digraph with no directed cycle. (10)

   (b) Prove that a graph is planar if and only if it has a dual. (10)

8. (a) Prove that a graph is embeddable in the plane if and only if it is embeddable on the sphere. (10)

   (b) Define max flow–min cut theorem. (10)

DE–512 **24**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2014.

PROGRAMMING IN C/C++

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Explain characters, integers and decimal numbers with examples. (10)

   (b) Discuss about 'data types' and functions in details. (10)

2. (a) Write a C++ program to check whether the given number is a palindrome or not. (10)

   (b) Explain the concept of object-oriented programming. (10)

3. (a) Explain all input and output commands in C and C++. (10)

   (b) Write a C++ program to convert Fahrenheit to Celsius. (10)

4. (a) Discuss about operators and its hierarchy in details with examples. (10)

   (b) Explain assignment operators and its shortcuts with examples. (10)

5. (a) What do you mean by control structures? Explain all kinds of control structures with examples. (10)

   (b) Discuss about goto, continue and break statements with examples. (10)

6. (a) Define Arrays and Initializations of arrays in details with examples. (10)

   (b) Write a C++ program to create ascending and descending order of 10 number using array. (10)

7. (a) Define pointers, functions and structures in details. (10)

   (b) Write a C program to print the given string in reserve order. (10)

8. (a) Explain input and output functions in file management with an example. (10)

   (b) Develop a C program to compute of two square matrices A and B of order n. (10)

DE–513 **25**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2014.

DISCRETE AND COMBINATORIAL MATHEMATICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) There are four partitions of 6 into odd summands : namely,

(1), (3), (6) and (10). We also have $P_d(6) = 4$. Is this a coincidence? If so explain why? (10)

(b) (i) Define '*generating function*'. (5)

(ii) Use generating functions to determine how many four-element subset of S contains no consecutive integers where $S = \{1, 2, ..., 15\}$. (5)

2. (a) (i) Define recurrence relation and linear recurrence relation. (5)

(ii) Find the recurrence relation for the Fibonacci sequence. (5)

(b) Solve the recurrence relation $a_{n+2} - 5a_{n+1} + 6a_n = 2$, $n \geq 0$, $a_0 = 3$, $a_1 = 7$. (10)

3. (a) Using the principle of inclusion and exclusion, prove that for $m \leq r \leq n$, $\binom{n \quad -m}{n \quad -r} = \sum_{i=0}^{m} (-1)^i \binom{m}{i} \binom{n-i}{r}$.

(10)

(b) A class consists of 12 boys and 5 girls. In how many ways a committee of five students can be formed if the committee is to consists of atleast one boy and one girl? (10)

4. (a) Explain principles of inclusion and exclusion as applied to three finite subjects. (10)

(b) Prove : Consider a set, with $|S| = N$ and conditions $c_i, 1 \le i \le t$, satisfied by some of the elements of S. Then the number of elements of S that satisfy none of the conditions $c_i, 1 \le i \le t$, be denoted by $\overline{N} = N(\overline{c}_1\ \overline{c}_2\ \overline{c}_3 ... \overline{c}_t)$ where, $\overline{N} = N - \sum_{1 \le i \le t} N(c_i) + \sum_{1 \le i < j \le t} N(c_i c_j) - \sum_{1 \le i < j < k \le t} N(c_i c_j c_k) + ... + (-1)^t N(c_1\ c_2\ c_3 ... c_t)$. (10)

5. (a) Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{u, v, w, x, y, z\}$, determine the number of one-to-one functions $f : A \to B$ where $f(1) \ne v, w; f(2) \ne u, w; f(3) \ne x$; and $f(4) \ne v, x, y$. (10)

(b) Explain Polya's theory of counting. (10)

6. (a) Let R be an equivalence relation on a finite set X. If each equivalence class has r elements then prove that there are $|X|^r$ equivalence classes. (10)

(b) Explain Polya's fundamental theorem. (10)

7. (a) Every finite Boolean algebra $\mathfrak{B}$ is isomorphic to a Boolean algebra of sets. (10)

(b) Prove that in a Boolean algebra $(L, \vee, \wedge)$ the complement $\overline{a}$ of any element $a \in L$ is unique. (10)

8. (a) Let $(L, \oplus *)$ be a Boolean lattice, then show that

(i) $\left(\overline{a \oplus b}\right) = \overline{a} * \overline{b}$

(ii) $\left(\overline{a * b}\right) = \overline{a} \oplus \overline{b}$. (10)

(b) Find a minimal-sum-of-products representation for the function $f(w, x, y, z) = \sum m(0, 1, 2, 3, 8, 9, 10)$. (10)